

12 The standard enthalpy change of combustion of but-1-ene, $\text{CH}_2=\text{CHCH}_2\text{CH}_3(\text{g})$, is $x \text{ kJ mol}^{-1}$.

The standard enthalpy change of the reaction $2\text{C}_2\text{H}_4(\text{g}) \longrightarrow \text{CH}_2=\text{CHCH}_2\text{CH}_3(\text{g})$ is $y \text{ kJ mol}^{-1}$.

What is the standard enthalpy change of combustion of ethene, $\text{C}_2\text{H}_4(\text{g})$?

A $\frac{x}{2} + y \text{ kJ mol}^{-1}$

B $x + \frac{y}{2} \text{ kJ mol}^{-1}$

C $\frac{x + y}{2} \text{ kJ mol}^{-1}$

D $\frac{x - y}{2} \text{ kJ mol}^{-1}$